Ann,. Rev. Psychol. 1993. 44:645–74

PROGRAM EVALUATION

Lee Sechrest and Aurelio José Figueredo

Evaluation Group for Analysis of Data, Department of Psychology, University of Arizona, Tucson, Arizona 85721

KEYWORDS: quantitative methods, qualitative methods, social ecology, Newtonian paradigm, Darwinian paradigm

CONTENTS

INTRODUCTION

Notable attempts have been made of late (e.g. Shadish et al 1991) to take stock of the accumulated experience of the last three decades of program evaluation in order to draw both theoretical and practical lessons for the present edification and future development of that enterprise. This discussion is in no way intended to detract from the numerous valuable insights provided by such recent historical analyses, but in the ultimately constructive spirit of multiplism (Shadish 1989), we wish to emphasize a partially complementary and perhaps convergent perspective.

We are concerned with the changing social context, or social ecology, of program evaluation. Shadish et al (1991), for example, have described the historical evolution of evaluation theory in at least two complementary ways: (*a*) a confrontation between an initial naiveté on the part of evaluation researchers and the unanticipated complexities of social and political reality, and

0066-4308/93/0201-645$02.00

(*b*) a partially cumulative progression and refinement of ideas, both theoretical and practical, within the field over that time. We do not challenge the usefulness of either of these perspectives, but propose the addition of other historical forces to the list. These additional and complementary interpretations are: (*a*) a confrontation between an initial physicalistic, or Newtonian, paradigm dominant in the social sciences and the ecological, or Darwinian, realities of the sociopolitical environment, and (*b*) a complex of real social changes during the historical development of program evaluation from an industrial era, or Taylorite, mode of management, in both business and government, to a postindustrial era, or cybernetic, mode of social intervention in what has increasingly become an information-based society. These two forces have altered not only how contemporary program evaluators relate to the outside world, but also what they conceive to be their special role within it.

Behavioral ecology is the scientific study of the relationships of the behavior of organisms to the relevant characteristics of the environment (Alcock 1989). It rests on the assumption that the behavior of living things is adapted, whether genetically or experientially, or both, to critical and identifiable features of their material conditions of existence (James 1910; Mayr 1974). We believe that this approach can shed some light on some of the historical trends that have been identified (e.g. by Shadish et al 1991) in the evolution of program evaluation. For example, consider the following questions. What accounts for the initial naiveté of evaluation researchers with respect to the unanticipated complexities of social and political reality? Why were they unanticipated? What accounts for the evolution of ideas within the program evaluation community? Is it merely the presumably inexorable forward march of scientific progress, or did some ideas become more salient and others less so over time because of changing social and political conditions?

Ecologically conscious program evaluators (e.g. Tharp & Gallimore 1979) have discussed the role of what they called evaluation pressure upon the serial succession of elements within social programs undergoing formative evaluations. We propose a selective ecological pressure of ambient social and political factors upon the evolving practice of program evaluation, which also undergoes formative development as a discipline. This pressure is contrary to the natural trend in self-consciously emerging fields to define themselves in terms of their unique characteristics and, thus, to mark off their own particular disciplinary boundaries. We intend to call attention to how some of the trends within the field of program evaluation have either paralleled or reflected trends outside of it.

WORLD-VIEWS IN COLLISION

Shadish et al (1991) marshal a convincing argument that program evaluation is not merely applied social science. Conceding that particular point, the question remains, why not? Shadish et al (1991) appeal to the peculiar problems manifest in program evaluation. However, these various problems arise not merely in program evaluation but whenever one tries to apply social science. The problems, then, arise not from the perverse peculiarities of program evaluation but from the manifest failure of much of mainstream social science and the identifiable reasons for that failure.

This failure of mainstream social science derives from the chronically inadequate external validity of the results of the dominant experimental research paradigm. Historically, this failure has been largely attributable to the nearly universal institutionalization of Fisherian null-hypothesis testing, as has been well documented elsewhere (e.g. Cohen 1990; Meehl 1978, 1990). In academic psychology, for example, the research paradigm originally established more than a century ago by Fechner, Weber, and Wundt (see Stigler 1986), which has exerted great influence to this day, was physicalistic, mechanistic, elementistic, and universalistic. This paradigm was at least partially inspired by a self-conscious desire to appear more scientific by identification with the indubitably more prestigious field of physics. As has also been documented elsewhere (Brunswik 1952, 1955; Petrinovich 1979, 1989), this physicalistic paradigm resulted in the incorporation of a model for both theory and research in psychology that has historically proven itself catastrophically inappropriate to a biological science (Mayr 1982).

The hallmark of this approach in the pragmatics of research was the ideal of the completely randomized and controlled single-variable experiment, or where possible, its logical extension, the fully crossed orthogonal factorial design. The supporting technology was the maximization of experimental variance, the minimization of error variance, and the control of extraneous variance, concisely summarized by the expression *maxmincon* (Kerlinger 1973). The hallmark of this approach in the construction of theory was a brute force inductivistic, automatized, and mechanistic search for ways to manipulate behavior systematically without the necessity for understanding it (Cohen 1990). Although the gritty realities of program evaluation fortunately prevented anyone from actually implementing these sterile procedures, the ideals of this research tradition were nonetheless quite influential. Program evaluation developed in the shadow of this reflected glory, as poor relations of the experimentalists. Indeed, even the contemporary literature on quasi-experimentation still shows this apologetic streak (e.g. Cook & Campbell 1979). Arguably, for quasi-experimentation, the more powerful and sophisticated intellectual engines of causal inference are superior, by now, to those of the experimental tradition.

The inappropriate research methodology, however, through its obvious impracticability in the field, did the least damage of all because it was quickly superseded by the methodological ingenuity of the quasi-experimentalists. The conceptual grip retained was a world-view based on the ideology of Newtonian space. The metaphor of a "social physics," as some early enthusiasts dubbed it (see Stigler 1986), is explicitly patterned on that of celestial mechanics and the dynamics of nonliving material particles. The resultant image is of a mechanical universe governed by absolute and unalterable laws, made of indivisible and identical units, of a finite number of specifiable types, majestically floating along predetermined pathways in a limitless void. In Newtonian space, bodies in motion tend to remain in motion, and bodies at rest tend to remain at rest unless acted upon by an external force.

Was it mere naiveté that accounted for the initial failure of evaluation researchers to anticipate the complexities of social and political reality? These researchers were mentally prepared by the dominant Newtonian paradigm of social science for a bold exploration of the icy depths of interplanetary space. Instead, they found themselves completely unprepared for the tropical nightmare of a Darwinian jungle: A steaming green Hell, where everything is alive and keenly aware of you, most things are venomous or poisonous or otherwise dangerous, and nothing waits passively to be acted upon by an external force. This complex world is viciously competitive and strategically unpredictable because information is power, and power confers competitive advantage. The Darwinian jungle manipulates and deceives the unwary wanderer into serving myriads of contrary and conflicting ends. The sweltering space suits just had to come off.

For a simple illustration of the complicating ecology of social programs, let us make a schematic path model out of the typical story of any programmatic intervention. We often start by observing, in the world, that a certain outcome, *O1,* seems to be produced by a specific determinant, *D1*. This outcome produces a negative utility, *U1,* to a specified population, so we implement a given intervention, *I1*. Thus, *I1* is intended to influence *D1,* to reduce *O1,* to minimize *U1*. That is straight-line Newtonian thinking. The problem is that *O1* has other determinants, *D2* and *D3,* which *I1* does not affect. Or, worse, *I1* does affect *D2* and *D3,* but in counterproductive ways. Furthermore, *D1* has other, unanticipated outcomes, *O2* and *O3,* which *D2* and *D3* might also affect in unknown ways. It also turns out that *O1* has different utilities, *U2* and *U3,* to different populations of stakeholders, as, perhaps, do *O2* and *O3*. What is a Newtonian to do? We might strategically design interventions *I2* and *I3,* to adequately regulate the system, but this requires ecological, i.e. Darwinian, thinking. This abstract scenario may sound a bit farfetched, but concrete and recent examples of all of these (and worse) complex contextual effects in program evaluation were described in a recent American Evaluation Association Panel on this subject (Scott & Figueredo 1991).

CONCURRENT SOCIAL REVOLUTION

To add to the litany of woes resulting from ecological complexity, the social ecology in which program evaluation must operate has been undergoing a process of radical transformation during the development of the field. This transformation has changed not only the fundamental organization of the society around them, but the specific niche that program evaluators occupy within it. Thus, what program evaluators have been learning about social ecology has been changing all around them as they work. As a result, many views of the mechanisms of social action that were held decades ago now seem quaint and unrealistic. Perhaps they were, but perhaps not. It is quite possible that many such notions were originally valid in the context of an industrial society. When the practice of program evaluation was officially initiated in the 1960s, the shift toward a postindustrial information-based economy, and corresponding ways of thinking, had just begun. The succession of dominant ideas within the field may reflect not just a progressive maturation of thinking through experience, but an evolved adaptation to a changing social environment.

For example, it has been claimed (Lincoln & Guba 1985) that the so-called rational model of social decision-making is now dead and, with it, the role of the program evaluator as an intelligence agent (or, at least, information provider) for the defunct rational policymakers. But what precisely does that mean? Does it mean that decisions are now somehow to be made irrationally? If so, precisely what kind of information is now required for the making of irrational decisions?

Rejecting this *reductio ad absurdum,* we do not think that anyone really believes that modern humans have somehow lost their rational faculties in planning complex social strategy. What is really meant is that the so-called "rational decisions" are no longer considered the exclusive privilege of government bureaucrats who paternalistically decide what is best for everyone. It may be that everyone, government bureaucrats included, continues to pursue rational self-interests in social relations. What has changed is that the designs, implementations, outcomes, and assigned utilities of social programs are no longer seen as necessarily serving any single set of interests, public or private, rational or otherwise. By implication, program evaluators are not exclusively beholden to any monolithic set of interests, objectives, or policymakers. Their clientele is the entire social network involved in and influenced by the social program, not just some specially empowered managerial hierarchy or elite. To call the interests and perspectives of all interested others irrational, by implication, reflects a transparent political ruse for continued domination by intellectual intimidation.

Another relevant change wrought by the postindustrial revolution is the blurring of the formerly clear distinction between producer and consumer. Because the postindustrial product is largely composed of information, feedback from the consumer has now become an integral part of the production process. As the distributor of that critical information, the program evaluator is now moving into an unprecedentedly powerful position in the postindustrial production process. The program evaluator, where fully integrated and utilized, has now literally become part of the social program itself, and not just an afterthought. The formative evaluation is an explicit example of this, but even the summative evaluation has become an implicit equivalent for future and related social programs. As a result of this and other social pressures, the Newtonian stance of complete impartiality has been explicitly abandoned by many program evaluators who have become active participants, and thus stakeholders in the social programs they evaluate. It would also be naive to think that program evaluators have not yet discovered any special interests of their own.

This diversity of interests, however, would be politically impotent were it not for the concomitant decentralization of the decision-making power and responsibility that characterizes postindustrial processes of production. This decentralization has all but made heterogeneity of treatment implementation a permanent fact of life. Indeed, many program evaluators no longer decry it as a failure of centralized program management and a threat to inferential validity, but rather exploit it as a productive source of experimental variance, which represents a potentially informative set of adaptations to and interactions with the heterogeneity of local molar conditions. With the centralization of program planning authority, the subsidiary doctrine of strict management accountability has also been lost because it has become nearly impossible to determine precisely who is to be held accountable for what. These economic and political developments are radically altering the social and ecological niche in which program evaluators operate. We propose that some of the changing practices observed in the discipline may at least partially reflect these pressures from a changing ambient social ecology.

EVALUATION IS NOT A FIELD

Neither program evaluation nor, as more broadly construed, evaluation, constitute a field of inquiry. Evaluation is an enterprise aimed at deciding the worth of various activities, and that enterprise comes equipped with a variety of assumptions and methods. As noted by Shadish et al (1991), we can evaluate anything. But if evaluation as an intellectual or professional activity is defined too broadly, it loses most of its usefulness in focusing attention and effort. A central set of issues exists with respect to program evaluation, and it is those we address in this review.

No cumulative body of substantive knowledge is attached specifically to evaluation. Thus, it would make no sense to try to review the findings of evaluation across fields of inquiry. For example, no more than idle curiosity would be satisfied by a review of programs that have been found effective in the past five years. Progress in evaluation is not to be measured by positive or negative findings any more than progress in statistics ought to be measured by the number of statistically significant findings reported in journals during some period. Progress in evaluation is to be measured in terms of advances in conceptualization of the enterprise and in improvements in methods for realizing its aims.

It is questionable whether program evaluation is cumulative, or at least whether cumulation of knowledge is proceeding with any rapidity. An investigation of the references in Shadish et al (1991), started at a more or less random point, revealed that of the next 130 references only 5 were dated 1988 or later; 58 were dated 1979 or earlier. A similar check of 130 references (starting at a different alphabetical point) in Chen (1990) showed that only 11 references were from 1987 or later; 75 were from 1979 or earlier. Checks of two current textbooks (Posavec & Carey 1989; Rossi & Freeman 1989) showed similarly small proportions of citations of recent material. Just by contrast, a check of a textbook in health psychology (Serafino 1990), picked more or less randomly from the shelf, showed that 37 of 130 references were dated 1987 or later, and only 26 were dated more than 10 years prior to the publication date of the book. We suspect that the dearth of recent citations in program evaluation reflects (*a*) the relative irrelevance of specific program evaluation outcomes, (*b*) the slow pace of development of new research designs and methods, and (*c*) lack of recognition of the important contributions from new developments in statistics and data analysis.

Results of evaluation studies, i.e. actual attempts to determine worth, are reported in virtually the full spectrum of social science journals. In fact, only a small proportion of evaluation findings could be reported in *Evaluation Review* (*ER*) or *New Directions in Program Evaluation.* Of 44 articles that appeared in Volume 4 (1980) of *ER*, only 15 had the primary purpose of reporting evaluation outcomes as opposed to proposing methodological improvements (24 articles) or clarifying concepts (4 articles). Still under the same editorship as in 1980, *ER* currently seems even less likely to contain articles whose primary purpose is to report outcomes of evaluation efforts, both because fewer, albeit somewhat longer articles, are being accepted and because whole issues are being devoted to special topics. In 5 numbers of Volume 15 of *ER* (Number 6 was a special issue on correcting of biases in voting rights cases) only two articles appeared to have as their primary purpose the reporting of an outcome of an evaluation; 15 articles were primarily methodological in nature, and 1 additional article concerned conditions of utilization of evaluation findings. *New Directions in Program Evaluation* also does not often publish the results of evaluations.

We conclude that current publication policies in core evaluation journals show a preference for methodological and conceptual articles and relegate evaluation reports either to specialized subject matter journals or to the fugitive literature represented by technical reports and government documents. One consequence is that most persons working in evaluation may only infrequently see a full-fledged evaluation report. Even worse, they may rarely see an exemplary report, a model to follow in their own work or to use in teaching and training activities.

The foregoing indicates that a useful review of evaluation must be directed at determining advances, if any, in conceptual frameworks and methods for evaluation. Reviews of evaluation findings are the provinces of individual fields in which the evaluations have been undertaken.

THE QUALITATIVE MOVEMENT

Evaluation began as an unabashedly quantitative enterprise. Early leaders in the field—D. T. Campbell, L. J. Cronbach, P. Rossi, R. J. Light, C. Weiss, for example—had in mind that evaluation would take advantage of the methods of research and analysis prevalent in the most prestigious domains of social science. It should be remembered, though, that in 1974 Donald T. Campbell gave an address at the American Psychological Association meetings entitled "On Qualitative Knowing in Action Research." That event was four years prior to the publication of Patton's (1978) *Utilization-Focused Evaluation,* arguably the first widely disseminated plea for qualitative evaluation, although Guba (1978) had a monograph in that same year. Patton's *Qualitative Evaluation Methods,* apparently the first textbook of its kind, was not published until 1980 (Patton 1980).

In the years since 1980, however, interest in qualitative approaches to evaluation has grown rapidly, perhaps sufficiently to justify identifying it as a movement. One index of that interest is that current membership in the Qualitative Topical Interest Group of the American Evaluation Association is now several times as large as that of the Quantitative Topical Interest Group, with only a small number of persons holding membership in both Groups (Sechrest 1992).

The question naturally arises why interest in qualitative approaches has developed so rapidly and so extensively. We do not believe that the answer can be found in any remarkable successes of qualitative evaluation. We have looked in vain for a qualitative evaluation that is widely cited as exemplary. We could find no instances of large social programs that have been qualitatively evaluated with national prominence given to the findings. (One reason for the latter may be, of course, that so few instances of large social programs can be found these days.) Qualitative evaluation seems confined to local efforts and to be directed more at improving ongoing programs than in determining their larger impact (e.g. Fetterman 1988, 1991). In his study of educational programs for gifted children Fetterman (1988) promises to describe "what does work" (p. xiv) in classrooms, but no criteria for how programs are to be judged are presented. For example, one cannot determine that children in the classes for the gifted are actually better off in the longer run for having been in them, let alone that other children in more ordinary classes are not harmed in any way.

Perhaps the qualitative movement can be seen as an alternative response to the social changes we have described, especially the changing roles and social context of the program evaluator. This variant response is evidenced by, among other things, the dissatisfaction expressed with the perceived limitations of traditional quantitative methods and the stronger emphasis placed upon multiple stakeholder interests and values. Unfortunately, the flight into qualitative methods can also be seen to reflect a reaction of despair at maintaining an analytic research program in the face of increasing social complexities. As will be detailed below, we believe that this reaction is unwarranted in the light of all the recent progress in the power and sophistication of multivariate quantitative models. Perhaps unlike the qualitative researchers, contemporary quantitative modelers might be up to the challenge of representing postmodern complexity without sacrificing analytical and scientific modes of thought.

If qualitative methods are ever going to be fully exploited, their methodologies must be better explicated, rationalized, and standardized. A critical property and advantage of many quantitative methodologies is that they are formulaic in nature. One need only follow the rules reasonably well in order to produce findings that have a reasonable probability of meeting with acceptance and of being repeatable. (This does not imply that competence is not an issue in quantitative evaluation, e.g. see Barkdoll 1992.) The same cannot be said for phenomenology, hermeneutics, critical analysis, and other qualitative methods. Their results appear likely to depend heavily on the outlook and skills of individual users. Whether ethnography should be considered standardized, or likely to become so, is open to question. Fetterman (1989) has made a useful effort toward standardization of ethnographic methodology for use in evaluation. Thus far, though, we agree with Chen (1990) that qualitative, or as he terms them, naturalistic, methods have not yet produced dependable and valid data.

Examples of the kinds of exegeses of qualitative methods that we consider exemplary are descriptions of grounded theory (Strauss & Corbin 1990) and case studies (General Accounting Office 1987). Whatever one might think about the ultimate usefulness of grounded theory methodology, Strauss & Corbin (1990) at least provide a comprehensive prescription for doing it. Grounded theory methodology is the most clearly formulated of the qualitative methods, and two conscientious investigators doing such studies would at least be able to communicate effectively the specific procedures followed and the stage of their work at any given time. Case studies may serve a variety of purposes in evaluation projects, but the methodology for doing them has been generally neglected (although see Yin 1984). Kazdin (1981) has made the case for applying the same questions about the validity of inferences from case studies as are usually asked about quasi-experiments, and the General Accounting Office (GAO) transfer paper (GAO 1987) provides an excellent framework within which to carry out most case studies. Qualitative research, like any other research, is worth doing well if it is worth doing at all.

Early discussions of the distinctions between quantitative and qualitative research in evaluation centered on their complementarity (Campbell 1974; Cook & Reichardt 1978). That is, the two approaches were seen as contributing different sorts of information bearing on the same general problem. Later works (e.g. Patton 1980, 1990) seemed to imply that qualitative methods might complement quantitative methods by focusing more on questions of process, such as whether interventions are properly implemented or whether results are likely to be differentially accepted by various stakeholder groups.

More recently, though, some writers have insisted that quantitative methods are completely outmoded and should be replaced by, not be complementary to, qualitative methods. These claims have been made especially insistently by Guba & Lincoln (1989) in their exposition of what they term "fourth generation evaluation." As yet no example of fourth generation evaluation seems to exist (at least not one so labeled); therefore one cannot tell exactly how such an evaluation might be carried out, or to what kinds of conclusions it might lead. Guba & Lincoln seem to believe, however, that highly satisfactory evaluations can be done without any intervention, any particular research design, any data collection (as it is ordinarily understood), or any form of quantitative analysis, let alone inferential statistics. They do intend, however, that the values of the investigator should be an integral feature of the evaluation although nothing in their book indicates that they recognize the possibility that some investigators might have politically and economically conservative values.

We believe that some proponents of qualitative methods have incorrectly framed the issue as an absolute either/or dichotomy. Many of the limitations that they attribute to quantitative methods have been discoursed upon extensively in the past. The distinction made previously, however, was not between quantitative and qualitative, but between exploratory and confirmatory research. This distinction is perhaps more useful because it represents the divergent properties of two complementary and sequential stages of the scientific process, rather than two alternative procedures. For example, Popper (1959) distinguished between the "context of discovery" and the "context of justification." In the context of discovery, free reign is given to speculative mental construction, creative thought, and subjective interpretation. In the context of justification, unfettered speculation is superseded by severe testing of formerly favored hypotheses, observance of a strict code of scientific objectivity, and the merciless exposure of one's theories to the gravest possible risk of falsification. That Popper clearly embraced both of these twin goals is evidenced by the slogan: "Boldness in conjecture, austerity in refutation." In reaction to what they perceive as an excess of the latter, some qualitative researchers appear to have become fixated at the earlier stage. Perhaps a compromise is possible in light of the realization that although rigorous theory testing is admittedly sterile and nonproductive without adequate theory development, creative theory construction is ultimately pointless without scientific verification.

Evaluations of any kind are done for purposes of influencing decisions and, in the case of social program evaluations, for influencing social policy. In the long run, the evaluations of evaluations must be in relation to their actual effect on social policy. Thus in the past 15 years there has been so much anguish over the question whether evaluation results are actually used in any way. (A Topical Interest Group in the American Evaluation Association, "Utilizing Evaluations," has even been formed around that question.) Thus, the fate of qualitative evaluation, including fourth generation evaluation, will lie in its persuasiveness. Of interest is that Strauss & Corbin (1990) specifically consider the credibility of grounded theory, and their criteria come down to, in our opinion, the credibility of the individual investigator.

What are qualitative evaluations like? The published works investigated are, for the most part, directed toward questions concerning values held by different stakeholders toward assumptions (to some extent theory) underlying interventions and toward implementation of interventions (process). Relatively little of the published material argues directly for conclusions about outcomes—did the program work or not? Qualitative evaluations are often surprisingly quantitative; it is just that their quantitative statements are imprecise. For example, instead of reporting that "55 percent of those interviewed said ...," the qualitative evaluation will simply say that "many of those interviewed said ... ." One of us (LS) was commenting to friends on his amusement that a report on a meeting of qualitative researchers persistently reported the exact number of persons in attendance at sessions. One listener noted that the difference is that, while a quantitative reporter would say "Only ten persons were present ...," a truly qualitative reporter would say, "Attendance at the session was depressing."

A case that illustrates some of the differences between the two camps is the differing conclusions about Jesse Jackson's program PUSH/Excel. A team was commissioned to do an evaluation of Jackson's program, which was receiving substantial federal funding. The relatively orthodox quantitative evaluation team came to distinctly negative conclusions about PUSH/Excel, not only because they could not find any evidence of consistently positive outcomes but because they could not identify a program. PUSH/Excel was promoted as a program with a set of central, organizing principles—a philosophy perhaps—meant to drive activities at many local sites. The evaluation team could not, however, find any consistent activities across sites; every site was different. Thus, the evaluators concluded that PUSH/Excel did not constitute a program suitable for federal funding. The leading critic of that evaluation was House (1988), a strong proponent of qualitative approaches to evaluation. The main thrust of House's critique, though, was not to produce counter evidence for the effectiveness of PUSH/Excel but to argue that the evaluators were wrong in insisting on trying to identify program elements consistent across sites in the first place. PUSH/Excel, in House's view, had to be allowed to develop variously to fit local conditions and resources; presumably it also had to be trusted to produce locally useful outcomes. What House never addressed in his critique was how Congress and others responsible for national policy and national funds were to decide about funding for programs that could not be specified and evaluated across sites. One cannot help but feel that the issues here, as in many other disagreements between evaluators of different persuasions, are matters of values rather than facts.

EVALUATIONS OF LARGE-SCALE SOCIAL PROGRAMS

Although evaluations of large-scale social programs do not seem as frequent as in the early years of program evaluation, the genre still exists. An example is the 1988 evaluation of a federal program to improve prospects of homeless persons for obtaining nutritious meals (Burt & Cohen 1988). Put simply, the intervention enabled homeless persons to use food stamps to purchase prepared meals from authorized providers. The evaluation plan required obtaining interview data on nationally representative samples of both homeless persons using the service and providers of the service. The samples were drawn from a nationally representative sample of 20 cities with populations of 100,000 or more; ultimately 1704 homeless persons and 381 providers were interviewed. In addition a sample of 142 homeless persons who had not used the meal service in the week prior to the interview were interviewed. Obviously, the study was meant to be more in the nature of a monitoring effort than an attempt to arrive at any penetrating causal inferences.

The research plan sounds simple, but the data needed were diverse and difficult to collect. The plan required data to be obtained on adequacy of meals in relation to both variety and nutritional content, on the sources of food served by providers, on the family status of the homeless persons, on their customary eating habits, on their use of food stamps, and many other issues. Presumably the fact that many (36%) of those persons using the meal service went at least one day per week with no food at all and that about a third of all the persons had had no contact at all with the food stamp program would have been useful to program planners. It is difficult to see how equally useful and persuasive information might have been obtained by other means, e.g. qualitative evaluation.

Large-scale national evaluation is not dead in the United States. Fourteen different programs to assess the potential value of providing housing services to homeless alcoholics are being implemented in thirteen cities across the nation (National Institute on Alcoholism and Alcohol Abuse 1990). Each of the separate programs has its own evaluation plan, but a national evaluation is also being carried out in an attempt to derive from all the separate programs and evaluations more general conclusions and recommendations. An important feature of the national evaluation is that each of the individual programs has been helped to develop a logic model for its efforts, i.e. a theory of the intervention (Lipsey 1990a). The logic models describe all the features of the intervention, whether they are mediating or moderating variables (Baron & Kenny 1986) that connect the planned treatments to the proposed outcomes. These models are then linked to data collection and analysis plans. Programs are encouraged to use some common measures and schedules for data collection. The national evaluation plan required the specification of an entirely separate research protocol, in fact, a separate design that will constitute a meta-analysis of individual program results. This monumental effort, sponsored by the National Institute on Alcoholism and Alcohol Abuse, will be worth following closely as it may constitute a model for national evaluations that could be widely emulated.

PROGRAM EVALUATION: THE FUNDAMENTAL ISSUES

Shadish et al (1991) provide a potentially useful conceptual scheme for thinking about approaches (theory) to program evaluation, but their ideas also provide a more general framework for carrying out program evaluations and for research directed at improving that enterprise. Five fundamental issues are at the heart of practical program evalution (Shadish et al 1991, p. 32):

1. Social programming: how social programs and policies develop, improve, and change, especially in regard to social problems;
2. Knowledge construction: how researchers learn about social action;
3. Valuing: how value can be attached to program descriptions;
4. Knowledge use: how social science information is used to modify programs and policies;
5. Evaluation practice: the tactics and strategies evaluators follow in their professional work, especially given their constraints.

These five points, in fact, encompass most of the activity in program evaluation today.

Social Programming and Theory-Driven Evaluation

The first issue is the focus on "theory driven evaluation" (Chen 1990; Chen & Rossi 1987, 1989). Many programs of social amelioration were launched with little thought to how they might actually have produced the expected effects. Moreover, when the programs were indifferently or even poorly implemented, as was almost always the case, no basis existed for determining the seriousness of departures from the original plans. In particular, Chen & Rossi advocate the explicit use of social science theory in order to devise, assess, and revise social programs. The longer range success of their prescription remains to be seen, but recent issues of *New Directions for Program Evaluation* have shown a distinct increase in emphasis on theory (e.g. Bickman 1990; Rog 1991), even when Chen & Rossi are not explicitly mentioned (e.g. Heilman 1991; Larson & Preskill 1991; Leviton et al 1990). Systematic application of social science theory could not help but be of some benefit both in the planning and evaluation of social programs, but for many social problems, e.g. drug addiction, crime, child abuse, domestic violence, the poverty of theory will be as evident as the theory of poverty.

Lipsey (1990a) argues strongly for more attention to theory of intervention as a complement to methodology. Under many circumstances, strong theory can compensate for relatively weak method. For example, strong theory can enable the development of more effective treatment, better measures, and more specific predictions about the exact circumstances under which effects are to be found. To argue, for example, that standardized outcome measures do not capture program effects (e.g. Hennessy & Grella 1992), is only helpful if alternative measures can be specified in advance. Theory helps to identify the focal and differential effects of treatments, saving both conceptual and statistical degrees of freedom.

Epistemology for Program Evaluation

The second critical issue raised by Shadish et al (1991), knowledge construction, is epistemological in nature: how do researchers obtain dependable knowledge about social programs. Perhaps it is an exaggeration to suggest that controversy rages, but the differences between those who urge rigorous and fundamentally quantitative approaches to evaluating social programs and those who urge more flexible and qualitative approaches are large and difficult to reconcile. As noted above, from the earliest years of program evaluation, it was assumed that both quantitative and qualitative information would be used in complementary ways to enhance understanding of programs and their effects. One has only to read the chapter titled "Approximations to knowledge" (Webb et al 1966), which was written early in the program evaluation movement, to have evidence of the commitment to multiple sources of knowledge. It is the recent insistence that qualitative evaluations can substitute for quantitative evaluations that is new (Guba & Lincoln 1989). This insistence is coupled with an increasing preoccupation with qualitative methods and their applications, and not accompanied by any concern at all for complementary quantitative data. None of three recent methodological books encouraging the use of qualitative methods for evaluation makes more than passing mention of any value of quantitative methods and data (Crabtree & Miller 1992; Fetterman 1989; Patton 1990).

Values and Evaluation

Valuing, the third issue listed by Shadish et al (1991), is related to another current and important controversy in program evaluation. Specifically, the issue is whether evaluation is, or ever could be, value free, and if it is not, whether evaluation should not operate openly in the service of particular values. Some writers (e.g. Guba & Lincoln 1989; House 1990; Sirotnik 1990) insist that evaluation can never be value free. Choice of perspectives, criteria, measures, methods, etc must necessarily reflect values. Therefore, these writers urge, evaluators should be partisan and carry out evaluations in the service of values. These writers also, however, seem to have a clear idea of which values should be served: Justice is the most frequently mentioned example.

Many conceptions of justice exist (Nozick 1974), but those writing about evaluation who insist on the explicit incorporation of values into the activity obviously have in mind a politically liberal version. Advocating the incorporation of values into program evaluation risks the very kind of partisanship that has aroused so much anger toward recent political regimes for their biased reviews of programs and policies.

It is not always easy to determine what the valuing controversy is all about. Early in the establishment of program evaluation as a formal enterprise, it was thought that evaluators would try to answer such questions as "Does this reading program improve reading skills?" or "Does this job training program help people get jobs?" Meta-questions such as "Why should these children be educated in a system that does not teach them to read by ordinary methods?" or "Where is the justice in a system that does not provide people with an adequate standard of living whatever their job status?" were scarcely imagined. The task of the evaluator was imagined rather like that of an appraiser who would give an objective estimate of the value of a piece of jewelry but who would not expect to be asked to give a special allowance for its sentimental value.

House (1990), to take but one example, seems to argue that the approach taken to the evaluation of two social programs, Follow Through and PUSH/Excel, were miscarriages of justice on largely sentimental grounds: The programs were for disadvantaged, largely minority populations and should have been valued for that reason. Nowhere in his critique of the evaluations does he provide any other basis for concluding that funding for the programs should have been continued, e.g. that they really were effective. Program evaluation may have to begin with values, but it will have to demonstrate objectivity if it is to retain any credibility at all.

Use of Program Evaluation Findings

The fourth issue raised by Shadish et al (1991) is knowledge use. The question whether program evaluation findings are used in any way once they are produced is one that has plagued the field since its beginning. For example, Weiss (1966) was writing about the problem well before the Evaluation Research Society was ever established. Early assumptions that evaluators would provide results of program evaluations to policymakers, who would then enact policies in accord with the results, now appear almost amusingly naive. On the other hand, that evaluation results would have had by this time such limited, irregular, and sometimes even perverse effects on social policy would likely have appalled the founders of program evaluation. Although such concepts as "enlightenment," proposed by Weiss (1977) and accepted by various other writers, have helped in understanding the complex relationships between evaluation results and public policy, it is still true that little can be stated with any confidence about the utilization of program evaluation findings, especially since much of the utilization that does occur may be conceptual rather than direct (Rossi & Freeman 1989).

Of special interest is the fact that most writing about utilization of program evaluation findings centers on public (usually national) policy. Very little is known about utilization of program evaluation results at state and local levels of government, and as far as we can tell, nothing at all may be known, at least systematically, about utilization of program evaluation results in the private sector. This is probably true in part because program evaluation in the private sector is proprietary and much of it is never published. Nonetheless, given that the private sector may be less political and somewhat less complex than the public policy sector, evaluation findings might have greater impact there. This is a prospect that merits examination and from which important lessons might be learned.

Evaluation Activities

The final issue mentioned by Shadish et al (1991) is evaluation practice: the specific activities carried out by program evaluators. Central to this issue, aside from the controversy over quantitative and qualitative approaches to evaluation, is formative versus summative evaluation. As originally conceived by Scriven (1967), formative and summative evaluation were two parts of the same process. Formative activities were those that guided program development to produce the best version possible of an intervention. Summative evaluation was then expected to follow in order to determine whether the best version possible had effects commensurate with original goals and with the costs and efforts required to produce the intervention. Summative evaluation was, though, hard to do. Determining whether programs worked required supporting a causal inference and, thus, a strong research design, careful attention to sample size and statistical power, and so on. Summative evaluation also required a commitment over a considerable period of time to the program and often to the evaluation itself. Finally, summative effects did not occur instantly. Unfortunately, evaluators and perhaps those commissioning their work have moved much in the direction of formative evaluation, which is, at least on a superficial level, easier. We admit that our assertion that evaluation has moved toward formative evaluation is a judgment, and in no way establishes that move with any certainty since published articles are a biased sample. We do not think it likely, though, that publication bias operates against summative evaluation. Three recent issues of *New Directions for Program Evaluation* aimed at evaluating training programs (Brinkerhoff 1989), AIDS prevention (Leviton et al 1990), and programs for the homeless (Rog 1991) do not contain a single article reporting outcomes of programs. This absence of material related to outcomes may be appropriate for the publication; nevertheless it is consistent with our point. Other evidence that leads us to believe that evaluators have shifted toward formative evaluations is the rapid growth of the Topical Interest Group on "Qualitative Methods" in the

AEA. Also an examination of the titles of papers from 12 randomly selected paper sessions of the 1991 meeting of the AEA, turned up only 12 of 51 titles that suggested that the papers might actually be reporting summative evaluation data.

We argued above that the ecology of evaluation has changed and, along with it, the roles and positions of evaluators and the nature of their activities. It is probably not in the nature of organizations and systems to seek summative evaluation of their own activities. The results of summative evaluation and even the rationale for doing it at all call into question the very reasons for existence of the organizations involved. Formative evaluation, by contrast, simply responds to the question, "How can we be better?" without strongly implying the question, "How do you know you are any good at all?" One has only to search for summative studies of university education, of which there are very few, to be reminded that even organizations with a tradition of intellectual inquiry have very little interest in evaluating themselves by their basic premises. Hence, if the relative prevalence of formative evaluation activities is increasing, relatively more evaluators may be employed within organizations that are commissioning the evaluations.

Still, we cannot escape the feeling that there has been, to some extent, a flight into qualitative, formative evaluation and away from the rigors of more quantitative, summative evaluation (Sechrest 1992). But we suspect also that focusing on formative evaluation helps to avoid the disappointments of summative evaluation. Rossi (1990) notes that, "Indeed, a good case can be made for the proposition that the expected value of the outcome of an impact assessment is zero or close to it" (p. 8). That expectation virtually guarantees that summative evaluation will be a discouraging activity. Even if formative evaluation is not easier to do, the findings are more comforting. Whatever else, one will at least be able to conclude that the program can be improved!

Program Development

In the 1970s, Tharp & Gallimore (1979) developed a model for program development, which they called the evaluation succession model. The model was seen as analogous to the biotic process (sere) by which a forest matures through a series of stages until finally reaching a climax stage of stability. Tharp & Gallimore postulated a similar progression for the development of effective social programs, and over a ten-year period they were able to construct a highly effective educational program for Hawaiian children. For example, their program increased reading scores of Hawaiian children from a very low level to a level about average for U.S. norms. Their program development process required use of the full array of research methods, data collection techniques, and so on. The process was iterative: At each step data from the previous step were fed back into the system to direct change and refinements. Although extensive use was made of the qualitative procedures, such as interviews and observations, and of formative evaluation approaches, summative (outcome) evaluations were required on a regular basis to determine the effectiveness of each successive version of the program. It is well known that the highly successful *Sesame Street* television program was developed by a similar iterative model of testing, feedback, and revision.

Unfortunately, the admirable model of Tharp & Gallimore appears to have gone unemulated. By and large, we are still at the level of either tinkering internally with programs by means of formative evaluation (without sufficient demonstration that the programs are effective) or conducting summative evaluations on one-shot programs (that are usually ineffective) and then proceeding to the next program. Why has the grand scheme of Tharp & Gallimore not caught on? First, their work is surprisingly little known in program evaluation, and awareness of the potentially far-reaching implications of their work is virtually zero. Two of the most recent textbooks in program evaluation (Posavac & Carey 1989; Rossi & Freeman 1989) do not even mention the Tharp & Gallimore project, although it remains one of the most successful intervention programs. The basic intervention program is proving highly effective in improving school performance of children in a Hispanic community in Los Angeles (R. Gallimore, personal communication, 1991). The second reason for the neglect of their model is more understandable: Their approach required ten years of continuing commitment and support from the funding agency and a matching commitment from the investigators. Few agencies, certainly not those of government, are in a position to make that kind of long term commitment and few investigators would be able and willing to stick with the same project for ten years, especially over the first several rather discouraging years. Still, if we are going to develop programs that are truly effective in our society, we must get beyond the notion of a quick fix, which is often implicit in our work. Tharp & Gallimore's funding agency, the Bishop Trust, in effect said, "We are going to solve this problem," not "Let's see if this idea works."

The essence of Tharp & Gallimore's evaluation succession is that one learns from one's mistakes. They started off their first year with what they thought was a reasonably good reading program. When that program proved a dismal failure, they did not set about finding another program to test. They asked themselves why what seemed like a good idea did not work at all, and they also asked other people, e.g. teachers, parents, even children. Then they built that feedback into the second year of their program, in the meantime also carrying out small-scale, controlled experiments of specific program elements. All over the country, shock incarceration is being tried as a way of getting youthful criminal offenders out of the criminal justice system and back into society. Predictably, it will prove disappointing in its outcomes, and equally predictably the response will be to close down the units and cast about for something else to do. Since shock incarceration is not a totally illogical intervention, and since it showed some promise in some places, a better response might be to try to figure out why it does not seem to work as well as expected and then, perhaps, try new versions of it, e.g. with better selection of candidates, better training of instructors, revised training schedule. Most of our social problems are of extreme difficulty. They will require commensurately strong efforts to deal with them, and program evaluation has, among its other important roles, the responsibility for giving policymakers the best information possible about the efforts that will be required.

Prospective Evaluation

Many children who are the offspring of disabled parents, e.g. drug addicts, mental patients, end up as public charges, although many others are taken in and cared for by relatives. A few years ago, it occurred to someone in New York that if more children were taken in by their relatives, the children would likely be better off and public expense for them would be reduced. Accordingly, a regulation was enacted that offered to pay relatives to take care of otherwise homeless children. The pay was fairly substantial but in line with responsibilities and savings to the public. In fairly short order the scheme collapsed because of extraordinarily high costs, which were way beyond what had been imagined. Suddenly people who had been taking care of children at their own expense wanted to be paid the same as anyone else. It is even likely that new cases were created when the natural mothers of children were forced out of their homes when it became apparent that their absence would result in a considerable increase in family income.

Could the result of the New York policy not have been foretold? In hindsight, exactly what happened seems an inescapable outcome. History is littered with stories of social interventions that did not have their intended effects or that had effects quite different from those intended, sometimes disastrous effects. The Program Evaluation and Methodology Division (PEMD) of the U.S. GAO is often asked by Congress to forecast the result of some program or policy change; they developed a process called "prospective evaluation" (GAO 1989c) to formalize the way they go about that task. The paper on prospective evaluation prepared by PEMD staff is a landmark in program evaluation. Much has yet to be learned about how to do prospective evaluations, but there is no reason to shirk the task. Prevention is better than cure, and if ways can be found to avert investments in ineffective or harmful programs, everyone will be better off.

New Evaluation Horizons

The title of this section is the same as the theme chosen for the 1991 meeting of the American Evaluation Association. That theme was meant to reflect the fact that the scope of evaluation has extended well beyond its original boundaries of public social programs. As noted, the concept of evaluation knows almost no bounds, and the methods employed are, for the most part, widely applicable and adaptable. Although not alone in doing so, the PEMD has been particularly notable for the increasing breadth of its evaluation efforts. The

USDA's commodity program, quality of the Federal workforce, the Paperwork Reduction Act, immigration policy, and accidental shootings have all been subjects of recent GAO/PEMD reports (GAO 1988a,b, 1989b, 1990c, 1991a). But PEMD has gone beyond even these areas to the evaluation of such physical systems as traffic congestion, hazardous waste management, highway safety, medical devices, and smart highways (GAO 1990b–e, 1990f, 1991b). A good bit of the work of GAO/PEMD is actually meta-evaluation, i.e. evaluation of the procedures by which other government agencies carry out their evaluations (GAO 1989c). Although GAO/PEMD is, perhaps, uniquely positioned to extend evaluation activities so greatly, other groups and persons could follow their lead, to the benefit of all concerned.

One of the consequences of extending evaluation to new areas is that one gets a sense of which issues and problems are general in nature and which may be limited to one or another area. Thus far, the issues and problems in evaluation appear much more general than specific. Proper specification of independent variables, construct validity of dependent variables, justifying causal inferences, and so on, are problems just as troubling in new areas as in the traditional ones of social services.

MULTIVARIATE METHODS IN PROGRAM EVALUATION

As noted elsewhere (Sechrest 1992), it is ironic that so much emphasis should now be placed on qualitative methods just when the development of quantitative methods is catching up to the demands of the complex tasks involved in program evaluation. Causal modeling with latent variables, growth curve analysis, regression-discontinuity analysis, and common factor measurement models are but a few examples of recent statistical developments or important improvements. These and other methods are now making it possible for data analysts to deal effectively with longitudinal data, unbalanced designs, multiple measures, and so on. Increased statistical sophistication, along with a much improved understanding of many of the basic design and logical issues put quantitative researchers in good position to respond to the challenges that qualitative methods are supposed to address.

It would be easy to exaggerate the influence of the philosophy of science on the everyday conduct of research, especially in program evaluation. By all the evidence, little of the research in program evaluation has been clearly guided by any particular philosophy of science. "Pointed in the right direction," might be about as far as one ought to go. Nevertheless, over the years the dominant philosophy of program evaluation has been Popper's, with its emphasis on falsification, i.e. the quest for residual truth. The time appears right for some reorientation.

Einhorn & Hogarth (1986) helped to clarify thinking about causality by showing the correspondence between everyday judgments of probable cause and both philosophical and empirical literature on the topic. A notable feature of their position is that causality becomes a positive rather than a residual inference. Cordray (1986) has elaborated on those ideas by suggesting that our usual criteria for positing causal relationships are impoverished. The situation is made worse by the inclination to treat quasi-experiments as only impoverished versions of true experiments rather than exploiting quasi-experiments fully for the information they do contain.

The change of emphasis advocated, then, is from a focus on ruling out rival causes or plausible rival hypotheses to a relatively stronger emphasis on ruling in causes, i.e. production of evidence and argument that makes our favored cause a plausible explanation for the data. In effect, this is what causal modeling attempts to do. The task of ruling in a cause may require casting our nets a bit wider for relevant evidence; it also puts more weight on theory. In the long run, though, we wish to generalize about causal relationships (Cook 1990), and these generalizations will best be supported by positive affirmation of the causal relationships. The good news is that increasingly powerful computers and statistical packages now offer researchers a wide range of statistical analyses capable of handling the large numbers of cases and variables required by an ecological systems approach. The bad news is that certain problems emerge from this cornucopia of statistical possibilities. Foremost among these is the possibility that researchers will end up using statistical analyses with little familiarity with their underlying assumptions and functional limitations. The use of unfamiliar statistical packages and procedures may, in turn, lead to inappropriate statistical analyses and unwarranted conclusions. The resulting problems can be very serious indeed, especially when conclusions are relevant to public policies. Another potential problem is that even technically competent researchers may become so immersed in and enamored of these new statistical techniques that their applications may cease to be the means to valid scientific ends and may instead become compulsive technological ends in themselves. One should avoid the technological imperative to apply them at every opportunity, regardless of the relative propriety of the occasion.

In order to support our strong claims with some concrete practical guidance, we review several of the more promising statistical options now available for the analysis of multiple predictors, multiple outcomes, multiple stakeholder utilities, and multiple independent replications.

Multiple Predictor Variables

Although widely accepted as a useful statistical tool, multiple regression/correlation analysis is fraught with dangers in estimating effect sizes when one uses a large number of predictor variables in the linear equation. For example, it is highly unlikely that a large number of naturally occurring predictors will be statistically independent. When two or more variables are relatively highly correlated, the statistical estimation method of squared error minimization used in multiple regression is incapable of sorting out their independent effects on the dependent variable. This condition is referred to as multicollinearity

(Pedhazur 1982) and results in highly unstable regression coefficients. Fortunately, useful diagnostic procedures for detecting multicollinearity exist (Pedhazur 1982; Cohen & Cohen 1983).

Because the least squares estimation procedure is unable to determine the independent effect of each variable, regression coefficients may be obtained that are actually: (*a*) higher than, (*b*) lower than, or (*c*) in a completely different direction from the bivariate correlation coefficient. Moreover, when correlations with another variable are nearly equal for two predictors, regression coefficients will be unstable across samples as first one predictor variable, then the other, happens to have the highest bivariate correlation (e.g. see Bingham et al 1991). Thus, correlated variables fed indiscriminately into a multiple regression equation may result in highly unstable and uninterpretable regression coefficients. Furthermore, serious possibilities of error lie in the interpretation of these unstable coefficients in regard to policy decisions (Figueredo et al 1991). Another substantial problem in using a large number of predictor variables in a multiple regression analysis is the possibility that coefficients may be reported as significant due to *alpha slippage*. Alpha slippage, sometimes known as capitalization on chance, can occur when testing large numbers of variables. The influence of multicollinearity and alpha slippage, although difficult to estimate, may cast considerable doubt on the validity and reliability of the final coefficients obtained by multiple regression.

Some researchers have fallen into the trap of using exploratory techniques of model specification, such as stepwise regression, to obtain a subset of predictors that can serve as proxies for the larger set through multicollinearity. The function of data reduction could be better served, however, not by using exploratory techniques of model specification, but by employing a common factor modeling technique. These multivariate methods capitalize on the problems of multiple regression by using the common variance, or multicollinearity, between predictor variables to construct hypothetical common factors, or latent variables, of which the measured variables are only manifest indicators. Hence, a large number of variables could be concisely represented by a few discriminable common factors; this would reduce the problem of alpha slippage and avoid the multicollinearity problems of multiple regression, while providing a parsimonious explanation of the phenomena. Another benefit of common factor modeling is that certain indicators that are clearly codetermined by more than one causal influence can be modeled as factorially complex, i.e. they indicate more than one hypothetical construct. By use of latent variable causal modeling, a factor analytic structural equation model can then be constructed in order to predict the outcomes of interest. At the very least, testing a common factor model can greatly facilitate understanding a complex data set. For instance, the inmate classification scale analyses reported by Bereccochea & Gibbs (1991) involved sets of 28 and 37 variables and a final set of 23 scorable items, developed by univariate tests and multiple logistic regressions. Because of weak theory and minimal data reduction, the final results are difficult to comprehend at any level of abstraction. Thus, an interpretable explanation need not be sacrificed for mere prediction as a goal of the analysis, and a conceptual virtue can be made out of a computational necessity.

Multiple Outcome Variables

In program evaluation, it is often the case that the multiple outcomes of a programmatic intervention need to be assessed. Moreover, these dependent variables are often selected to measure the impact of the intervention on several conceptually distinct outcomes rather than converge upon a single construct. Thus, multivariate data reduction methods, such as common factor modeling, may not be appropriate. Even where common factor modeling is appropriate, there remains the question how then to deal with the multiple dependent common factors that one has constructed. Because the programmatic intervention may exert a common causal influence on these multiple outcomes, whether manifest or latent, the dependent variables are likely to be at least spuriously correlated with each other. In addition, the multiple outcomes may also subsequently exert various causal influences on each other. Thus, separate causal analyses for each of these dependent variables may also not be appropriate.

Several statistical procedures are available for the analysis of multiple correlated dependent variables. One of these is structural equations modeling, or confirmatory path analysis, in which the hypothesized causal network between outcomes can be fully specified, estimated, and tested. This method, however, requires the guidance of a strong causal theory, which is often not available in program evaluation. Another multivariate method is simultaneous canonical analysis, which requires little theory. This method, however, produces empirically derived linear composites of dependent variables, or canonical variates, which are almost always difficult to interpret pragmatically in terms of concrete outcomes.

A third alternative is sequential canonical analysis, which combines some of the advantages of the previous two procedures. Rather than combining all dependent variables into uninterpretable linear composites, as in simultaneous regression of independent variables, it partitions their covariance sequentially, as in hierarchical regression, while maintaining their separate identity. This method isolates the direct effects of the independent variables or interventions sequentially on each of the dependent variables or outcomes, controlling for all indirect effects through the prior dependent variables. Because the only theoretical guidance required is a tentative specification of the causal order between the dependent variables, this sequential method is currently being developed into an exploratory form of path analysis (Gorsuch & Figueredo 1991).

One example of a potential application of sequential canonical analysis is the study of a juvenile justice program (Land et al 1990), which involved three outcome variables: delinquent offenses, status offenses, and judged overall success of the case. In the reported study, these three outcomes were separately analyzed as univariate indicators. A sequential canonical analysis would perhaps have determined first whether the intervention affected status offenses, then whether there was a significant residual effect on delinquent offenses after adjusting for status offense differences, and then whether judged success had a significant residual after removing any effect on delinquent offenses.

Multiple Utility Functions

In contrast to estimates of the relative effectiveness of programmatic interventions in producing specified objective outcomes, the estimates of the relative subjective utilities of those outcomes often used are only sometimes empirical in derivation, and then in varying degrees. One common index of utility is monetary cost (Kaplan 1985). Whereas few would recommend relying exclusively on money as the sole measure of utility, and thus (by default) the operational definition, most would recognize it as one important element in a multiple operationalization. Often, any nonmonetary outcome utilities are arbitrarily assigned, as when there is a clear and uncontroversial choice between outcomes with utilities of 1 and 0, such as life and death. Perhaps more often they are developed by consensus among a panel of expert judges. Technically, these judgments qualify as somewhat empirical estimates, but they may fail to hold up under critical scrutiny of methodological issues, such as parametric generalizability beyond the group of panel members sampled.

Quantifying utilities is the area of social decision-making that is least sophisticated in either methodological rigor or in just plain clear thinking about the nature of the problem. For example, assuming that adequate techniques are available for sampling utilities from a duly constituted representative sample, what precisely is the population of interest, or intended universe of generalization? One possible solution that has been suggested to this dilemma is to specify precisely what utilities are being estimated and what decision rules are being applied. A critical multiplist (Shadish 1989) alternative would be to simulate model predictions separately using the utilities obtained from different subject populations. Such substitution of values would constitute a kind of representative sensitivity analysis to the issue of what set of stakeholder interests are being considered. If they are to be applied as influential coefficients in our quantitative models, we need to know just what those numbers mean. If we do not know precisely what our assigned utilities represent, the whole result becomes uninterpretable.

Even with clear thinking on such thorny issues, there remain a number of quite formidable methodological obstacles to the very measurement of subjective utility. Unfortunately, psychological research has shown that many people are notoriously poor judges of numerical probability. Also, the alternatives posed in utility estimation are often also hypothetical: Subjects have had no direct experience with the outcomes in question. Another interesting feature of the current use of expected utility theory is that it appears to have been applied exclusively to the evaluation of discrete (e.g. binary) outcomes. There is, however, no immediately obvious mathematical reason to preclude its application to continuously graded outcomes. A certain degree of graded effect could be proportionately evaluated by a utility coefficient in much the same way that an all-or-nothing dichotomy is evaluated. Since the probabilities of discrete outcomes can be modeled as mere special cases of regression coefficients for which the criterion variables are dichotomous (Cohen & Cohen 1983), effect sizes from multiple regression analyses in general could perhaps be used to construct optimality models in which the graded levels of the dependent variables are weighted in such utility functions.

The graded equivalent of a discrete decision tree is a path model in which the multiplication of successive path regression coefficients in the computation of indirect effects is the exact functional equivalent of that of a series of Bayesian conditional probabilities. Utility coefficients could be represented as additional causal pathways to a single utility construct from all the dependent variables. This kind of continuous information might be even more difficult to obtain than the utilities of discrete outcomes. However, since a substantial proportion of the available parameter estimates of the effectiveness of means of intervention are reported in a continuous, rather than a discrete, metric, this adaptation would allow a more powerful and potentially widespread application of the benefits claimed for expected utility theory. The use of such decision-analytic path models is described more fully elsewhere (Sechrest & Figueredo 1992).

Despite the considerable attention paid to values in the literature on program evaluation, the concept of utility is almost completely missing. The notion of utility is, however, scarcely ever far from the surface of consciousness in evaluation studies. Utility refers to whatever comes after the immediate outcome of some intervention in terms of its impact on the lives of persons, organizations, communities. A drug to cure septicemia might well preserve many lives, but if many of those persons would be likely to die soon anyway of other problems, the drug might be regarded as successful but as having little utility. Suppose one developed a program that would keep some youth from dropping out of high school. But suppose one found that those students would not really be better off in any particular way for having finished high school. The latter involves the concept of utility. Few evaluation studies assess the utility of the programs that are the focus of interest.

Benefit-cost studies almost always seem to have disappointing results because the amount of money saved is scarcely ever more than marginally larger than the money spent. We would like to spend \$1 on programs and get a return of \$10. A more common result is \$1.25. The reason for the low return is that monetized benefits tend to reflect utilities, the step(s) beyond immediate outcomes. That step represents an additional path in the path diagram and, hence, an extra multiplication. Even if the paths from intervention to outcome multiply out to .5, which would be a rather strong relationship, and if the relationship between outcome and utility is only .5, again a fairly strong relationship, then the overall path has a value of only .25. Put another way, variance in outcome accounted for of .25 is reduced to .06 variance in utility accounted for. Program evaluators may not like utility analyses when they get around to doing them.

Multiple Independent Replications

Meta-analysis of data sets is increasing in frequency and in influence on the field. The use of meta-analysis in program evaluation is likely to be somewhat restricted because of the relatively small number of programs that are ever tested in multiple sites and studies. Meta-analyses of medical and even psychological therapies often involve dozens of studies. Few social programs are likely to be tested so diversely. Too often the versions of interventions that are tested vary so greatly in almost all characteristics as to defy aggregation for purposes of meta-analysis. A partial exception is the ongoing multi-site study of programs for the homeless (National Institute for Alcohol Abuse and Alcoholism 1991). At least for those programs agreements were reached for some standardization of measures and for some common points of testing. A recent GAO report (GAO 1992) of unusual interest and potential importance offers new perspectives on synthesis of data across studies with different designs. In the long run, meta-analysis is likely to prove a good bit more useful for interpreting bodies of literature, e.g. for understanding our research and limitations on its generalizability, than for estimating overall effect sizes (Cordray 1990; Lipsey et al 1986).

One instructive meta-analysis (although it is nonquantitative) is the volume by Bloom et al (1988), for which they requested six papers summarizing evaluations of programs in six social problem areas. The editors then synthesized the lessons from across the six areas to produce the following five conclusions:

1. Effects of programs or policies are likely to be small but nonetheless may be quite important.
2. Treatments—even those that carry the same label—vary considerably. Looking inside the black box of treatment will help us to understand why programs succeed or not.
3. The effects of treatments are likly to vary substantially across the different target groups and locales.
4. Evaluations are often limited by small sample sizes, the implementation of weak and variable treatments, inadequate outcome variables, and poor research designs. These problems can be circumvented in many cases.
5. Results of individual, well executed evaluations contribute to our cumulative understanding of program theory, process, and outcomes.

Those five points sum up the current state of our knowledge of program evaluation.

Sechrest et al (1979) insisted quite some time ago that more attention should be paid to the concepts of strength and integrity of treatment and to their quantification. Although this idea is not yet ubiquitous, it may be catching on (e.g. see Chen 1990; Posavac & Carey 1989). Dennis and his colleagues (e.g. Dennis et al 1991) have been especially attentive to issues of strength and integrity of treatment in their evaluations of drug treatment programs. Quantification of strengths of social interventions would have the advantage of providing an important extra item of information in statistical analyses; intervention could be coded as a continuous rather than a binary variable. That would under many circumstances bring into play the concept of the dose-response relationship in outcome evaluations (Lipsey 1990). A second advantage of quantification of interventions would be much needed protection against Type 2 error in statistical conclusion validity. Low statistical power is as much a function of effect size as of sample size (Lipsey 1990), and if treatment strength were known to be low, failure to find an effect would not be as likely to be mistakenly attributed to ineffectiveness of the intervention.

RECENT DEVELOPMENTS IN THE PROFESSION OF PROGRAM EVALUATION

Just about the time of the last review in this series of program evaluation (Shadish & Cook 1986), the former Evaluation Research Society (ERS) and the Evaluation Network (ENet) merged to form the American Evaluation Association. That merger was not achieved without difficulty since it brought together a group (ERS) of predominantly academic and quantitative researchers with a group (ENet) often referred to as practitioners who are more interested in doing evaluation than in worrying about its theory and methods. A second signal event was the demise of the *Evaluation Studies Review Annual* (*ESRA*) with Volume 12 (Shadish & Reichardt 1987). Perhaps the ERS/ENet merger and the end of the *ESRA* are linked. *ESRA* was always a formidable volume that unquestionably appealed more to those evaluators with theoretical and methodological interests than to those with more pragmatic concerns. Indeed, it served as yearbook for the field (Shadish et al 1991), and the fact that it could not survive is not grounds for optimism about the future of the field. The AEA, itself, has languished to some extent, its membership hovering around the 2500 mark since its founding. Lack of growth in membership appears to reflect less the failure to recruit new members each year than the failure to retain the loyalties of enough of its members.

On the other hand, the AEA, in concert with the Canadian Evaluation Society, is planning the First International Congress on Evaluation, to take place in Vancouver, BC in 1995. Perhaps that international event may have a rejuvenating effect on the field.

Literature Cited

Alcock, J. 1989. *Animal Behavior: An Evolutionary Approach.* Sunderland, Mass: Sinauer.

Barkdoll, G. L. 1992. Strong medicine and unintended consequences. *Eval. Pract.* 13:53–57

Baron, R. M., Kenny, D. A. 1986. The moderator-mediator variable distinction in social psychological research: conceptual, strategic, and statistical considerations. *J. Pers. Soc. Psychol.* 51:1173–82

Bereccochea, J. E., Gibbs, J. B. 1991. Inmate classification: a program that works? *Eval. Rev.* 15:333–63

Bickman, L., ed. 1990. *Advances in Program Theory. New Directions for Program Evaluation,* No. 47. San Francisco: Jossey-Bass.

Bingham, R. D., Heywood, J. S., White, S. B. 1991. Evaluating schools and teachers based on student performance: testing an alternative methodology. *Eval. Rev.* 15:191–218

Bloom, H. S., Cordray, D. S., Light, R. J., eds. 1988. *Lessons from Selected Program and Policy Areas. New Directions for Program Evaluation,* No. 37. San Francisco: Jossey-Bass.

Brinkerhoff, R. O., ed. 1989. *Evaluating Training Programs in Business and Industry. New Directions for Program Evaluation,* No. 44. San Francisco: Jossey-Bass

Brunswik, E. 1952. The conceptual framework of psychology. In *International Encyclopedia of Unified Science,* ed. O. Neurath, R. Carnap, C. Morris, 1:655–760. Univ. Chicago Press

Brunswik, E. 1955. Representative design and probabilistic theory in a functional psychology. *Psychol. Rev.* 2:193–217

Burt, M. R., Cohen, B. E. 1988. *Feeding the homeless: does the prepared meals provision help? Rep. Congr. Prep. Meal Prov.,* Vol. 1. Washington, DC: The Urban Inst.

Campbell, D. T. 1974. On qualitative knowing in action research. In *Methodology and Epistemology for Social Science: Selected Papers of Donald T. Campbell,* ed. E. S. Overman, pp. 360–76. Chicago: Univ. Chicago Press

Chen, H. 1990. *Theory-Driven Evaluations.* Newbury Park, Calif: Sage. 336 pp,

Chen, H., Rossi, P. H. 1987. The theory-driven approach to validity. *Eval. Program Plan.* 10:95–103

Chen, H., Rossi, P. H. 1989. Issues in the theory-driven perspective. *Eval. Program Plan.* 12:299–306

Cohen, J. 1990. Things I have learned (so far). *Am. Psychol.* 45:1304–12

Cohen, J., Cohen, P. 1983. *Applied Multiple Regression/Correlation Analysis for the Behavioral Sciences.* Hillsdale, NJ: Erlbaum

Cook, T. D. 1990. The generalization of causal connections: multiple theories in search of clear practice. In *Research Methodology: Strengthening Causal Interpretations of Nonexperimental Data,* ed. L. Sechrest, E. Perris, J. Bunker, pp. 9–32. Rockville, MD: Agency Health Care Policy Res.

Cook, T. D., Campbell, D. T. 1979. *Quasi-Experimentation: Design and Analysis Issues for Field Settings.* Boston, Mass: Houghton Mifflin

Cordray, D. S. 1990. Strengthening causal interpretations of nonexperimental data: the role of meta-analysis. In *Research Methodology: Strengthening Causal Interpretations of Nonexperimental Data,* ed. L. Sechrest, E. Perrin, J. Bunker, pp. 151–72. Rockville, Md: Agency for Health Care Policy Res.

Crabtree, B. F., Miller, W. L., eds. 1992. *Doing Qualitative Research,* Vol. 3,*Research Methods for Primary Care.* Newbury Park, Calif: Sage

Dennis, M. L., Fairbank, J. A., Bonito, A. J., Rachal, J. V. 1991. *Treatment Process Study Design. Methadone Enhanced Treatment (MET) Trials Tech. Doc. 7.* Research Triangle Park, NC: Research Triangle Inst.

Einhorn, H. J., Hogarth, R. M. 1986. Judging probable cause. *Psychol. Bull.* 9:3–19

Fetterman, D. M. 1988. *Excellence and Equality: A Qualitatively Different Perspective on Gifted and Talented Education.* Albany: State Univ. New York Press

Fetterman, D. M. 1989. *Ethnography: Step by Step.* Newbury Park, Calif: Sage

Fetterman, D. M., ed. 1991. *Using Qualitative Methods in Institutional Research. New Directions for Institutional Research,* No. 72. San Francisco: Jossey Bass

Figueredo, A. J., Hetherington, J., Sechrest, L. 1991. Water under the bridge: A response to Bingham, Heywood, and White. *Eval. Rev.* 17:40–62

General Accounting Office 1987. *Case Study Evaluations. GAO/PEMD-87-9.* Washington, DC: US Gen. Account. Off.

General Accounting Office 1988a. *USDA's Commodity Program: The Accuracy of Budget Forecasts. GAO/PEMD-88-8.* Washington, DC: US Gen. Account. Off.

General Accounting Office 1988b. *Federal Workforce: A Framework for Studying Its Quality Over Time. GAO/PEMD-88-27.* Washington, DC: US Gen. Account. Off.

General Accounting Office 1989a. *Medical Devices: FDA's Implementation of the Medical Device Reporting Regulation. GAO/PEMD-89-10.* Washington, DC: US Gen. Account. Off.

General Accounting Office 1989b. *Paperwork Reduction: Mixed Effects on Agency Deci-*

sion Processes and Data Availability. GAO/PEMD-89-20. Washington, DC: US Gen. Account. Off.

General Accounting Office 1989c. *Prospective Evaluation Methods: The Prospective Evaluation Synthesis. GAO/PEMD-89-10.* Washington, DC: US Gen. Account. Off.

General Accounting Office 1990a. *Traffic Congestion: Federal Efforts to Improve Mobility. GAO/PEMD-90-2.* Washington, DC: US Gen. Account. Off.

General Accounting Office 1990b. *Hazardous Waste: EPA's Generation and Management Data Need Further Improvement. GAO/PEMD-90-3.* Washington, DC: US Gen. Account. Off.

General Accounting Office 1990c. *Immigration Reform: Major Changes Likely Under S358. GAO/PEMD-90-5.* Washington, DC: US Gen. Account. Off.

General Accounting Office 1990d. *Medical Device Recalls: Examination of Selected Cases. GAO/PEMD-90-6.* Washington, DC: US Gen. Account. Off.

General Accounting Office 1990e. *Highway Safety: Trends in Highway Fatalities 1975-87. GAO/PEMD-90-10.* Washington, DC: US Gen. Account. Off.

General Accounting Office 1991a. *Accidental Shootings: Many Deaths and Injuries Caused by Firearms Could Be Prevented. GAO/PEMD-91-9.* Washington, DC: US Gen. Account. Off.

General Accounting Office 1991b. *Smart Highways: An Assessment of Their Potential to Improve Travel. GAO/PEMD-91-18.* Washington, DC: US Gen. Account. Off.

General Accounting Office 1992. *Cross Design Synthesis: A New Strategy for Medical Effectiveness Research. GAO/-PEMD-92-18.* Washington, DC: US Gen. Account Off.

Gorsuch, R. L., Figueredo, A. J. 1991. *Sequential canonical analysis as an exploratory form of path analysis.* Presented at Am. Eval. Assoc. Conf.: New Eval. Horizons. Chicago

Guba, E. G., Lincoln, Y. S. 1989. *Fourth Generation Evaluation.* Newbury Park, Calif: Sage

Heilman, J. G., ed. 1991. *Evaluation and Privatization: Cases in Waste Management. New Directions for Program Evaluation,* No. 51. San Francisco: Jossey-Bass

Hennessy, M., Grella, C. 1992. Evaluating "innovative" programs for homeless persons: case study of an unsuccessful proposal. *Eval. Pract.* 13:15–25

House, E. R. 1988. *Jesse Jackson and the Politics of Charisma: The Rise and Fall of the PUSH/Excel Program.* Boulder, Colo: Westview

House, E. R. 1990. Methodology and justice. See Sirotnik 1990, pp. 23–36

James, W. 1910. *Principles of Psychology.* Vol. I. New York: Holt

Kaplan, R. M. 1985. Quantification of health outcomes for policy studies in behavioral epidemiology. In *Behavioral Epidemiology and Disease Prevention,* ed. R. M. Kaplan, M. H. Criqui, pp. 31–54. New York: Plenum

Kazdin, A. E. 1981. Drawing valid inferences from case studies. *J. Consult. Clin. Psychol.* 49:183–92

Kerlinger, F. N. 1973. *Foundations of Behavioral Research.* New York: Holt, Rinehart & Winston

Land, K. C., McCall, P. L., Williams, J. R. 1990. Something that works in juvenile justice: an evaluation of the North Carolina Court Counselors' Intensive Protective Supervision Randomized Experimental Project, 1987–89. *Eval. Rev.* 14:574–606

Larson, C. L., Preskill, H., eds. 1991. *Organizations in Transition: Opportunities and Challenges for Evaluation. New Directions for Program Evaluation,* No. 49. San Francisco: Jossey-Bass

Leviton, L. C., Hegedus, A. M., Kubrin, A., eds. 1990. *Evaluating AIDS Prevention: Contributions of Multiple Disciplines. New Directions for Program Evaluation,* No. 46. San Francisco: Jossey-Bass

Lincoln, Y. S., Guba, E. G. 1985. *Naturalistic Inquiry.* Beverly Hills, CA: Sage

Lipsey, M. W. 1990a. Theory as method: small theories of treatments. In *Research Methodology: Strengthening Causal Interpretations of Nonexperimental Data,* pp. 33–52. Rockville, MD: Agency Health Care Policy Res.

Lipsey, M. W. 1990b. *Design Sensitivity: Statistical Power for Experimental Research.* Newbury Park, Calif: Sage

Lipsey, M. W., Crosse, S., Dunkle, J., Pollard, J., Stobart, G. 1986. Evaluation: the state of the art and the sorry state of the science. In *Evaluation Studies Review Annual,* ed. D. S. Cordray, M. W. Lipsey, 10:153–74. Beverly Hills: Sage

Mayr, E. 1974. Behavioral programs and evolutionary strategies. *Am. Sci.* 62:650–59

Mayr, E. 1982. *The Growth of Biological Thought: Diversity, Evolution, and Inheritance.* Cambridge, Mass: Harvard Univ. Press

Meehl, P. E. 1978. Theoretical risks and tabular asterisks: Sir Karl, Sir Ronald, and the slow progress of soft psychology. *J. Consult. Clin. Psychol.* 46:806–34

National Institute on Alcohol Abuse and Alcoholism. 1990. *Cooperative agreements for research demonstration projects on alcohol and other drug abuse treatment for homeless persons. RFA LA-90-01.* Washington, DC: Public Health Serv.

National Institute on Alcohol Abuse and Alcoholism. 1991. *National Evaluation Plan: Homelessness and Alcohol/Drug Abuse.* Rockville, MD: Natl. Inst. Alcohol Abuse and Alcoholism

Nozick, R. 1974. *Anarchy, State, and Utopia.* New York: Basic Books

Patton, M. Q. 1978. *Utilization-Focused Evaluation.* Beverly Hills, CA: Sage

Patton, M. Q. 1980. *Qualitative Evaluation Methods.* Beverly Hills, Calif: Sage

Patton, M. Q. 1990. *Qualitative Evaluation Methods and Research Methods.* Newbury Park, Calif: Sage

Pedhazur, E. J. 1982. *Multiple Regression in Behavioral Research: Explanation and Prediction.* New York: Holt, Rinehart & Winston

Petrinovich, L. 1979. Probabilistic functionalism: A conception of research method. *Am. Psychol.* 34:373–90

Petrinovich, L. 1989. Representative design and the quality of generalization. In *Everyday Cognition in Adulthood and Late Life,* ed. L. W. Poon, D. C. Rubin, B. A. Wilson, pp. 11–24. New York: Cambridge Univ. Press

Popper, K. 1959. *The Logic of Scientific Discovery.* New York: Basic Books. 479 pp.

Rog, D. J., ed. 1991. *Evaluating Programs for the Homeless. New Directions for Program Evaluation,* No. 52. San Francisco: Jossey-Bass

Rossi, P. H., Freeman, H. E. 1989. *Evaluation: A Systematic Approach.* Newbury Park, CA: Sage

Rossi, P. H. 1990. Foreword. See Chen 1990, pp. 7–10

Scott, A. G., Figueredo, A. J. 1991. Evaluating the ecological impact of social interventions. Discussed at Am. Eval. Assoc. Conf.: New Eval. Horizons. Chicago.

Scriven, M. 1967. The methodology of evaluation. In *Perspectives of Curriculum Evaluation,* ed. R. M. Tyler, R. M. Gagne, M. Scriven, pp. 39–83. Chicago: McNally

Sechrest, L. 1992. Roots: back to our first generations. *Eval. Pract.* 13:1–7

Sechrest, L., Figueredo, A. J. 1992. Approaches used in conducting outcomes and effectiveness research. In *A Research Agenda for Outcomes and Effectiveness Research,* ed. P. Budetti. Alexandria, Va: Health Adm. Press. In press

Sechrest, L., West, S. G., Phillips, M. A., Redner, R., Yeaton, W. 1979. Some neglected problems in evaluation research: strength and integrity of treatments. In *Evaluation Studies Review Annual,* ed. L. Sechrest, S. G. West, M. A. Phillips, R. Redner, W. Yeaton, 4:15–35. Beverly Hills, Calif: Sage

Serafino, E. P. 1990. *Health Psychology: Biopsychosocial Interaction* . New York: John Wiley

Shadish, W. R. Jr. 1989. Critical multiplism: a research strategy and its attendant tactics. In *Health Services Research: A Focus on AIDS,* ed. L. Sechrest, H. Freeman, A. Mulley, pp. 5–28. Rockville, MD: Agency Health Care Policy Res.

Shadish, W. R. Jr., Cook, T. D. 1986. Program evaluation: the worldly science. *Annu. Rev. Psychol.* 37:193–232

Shadish, W. R. Jr., Cook, T. D., Leviton, L. C. 1991. *Foundations of Program Evaluation: Theories of Practice.* Newbury Park, Calif: Sage

Shadish, W. R. Jr., Reichardt, C. S. 1987. *The Evaluation Studies Review Annual,* Vol. 12. Newbury Park, Calif: Sage

Sirotnik, K. A. 1990. Evaluation as critical inquiry: school improvement as a case in point. In *Evaluation and Social Justice: Issues in Public Education. New Directions for Program Evaluation,* No. 45, ed. K. A. Sirotnik, pp. 37–60. Newbury Park, Calif: Sage

Stigler, S. M. 1986. *The History of Statistics: The Measurement of Uncertainty Before 1900.* Cambridge, Mass: Harvard Univ. Press

Strauss, A., Corbin, J. 1990. *Basics of Qualitative Research: Grounded Theory Procedures and Techniques.* Newbury Park, Calif: Sage

Tharp, R. G., Gallimore, R. 1979. The ecology of program research and evaluation: A model of evaluation succession. See Sechrest et al 1979, pp. 39–60

Webb, E., Campbell, D. T., Schwartz, R. D., Sechrest, L. 1966. *Unobtrusive Measures: Nonreactive Measures in the Social Sciences.* Chicago: McNally

Weiss, C. H. 1966. Utilization of evaluation: toward comparative study. Paper presented at Am. Sociol. Assoc., Miami Beach

Weiss, C. H. 1977. Research for policy's sake: the enlightenment function of social research. *Policy Anal.* 3:531–45

Yin, R. K. 1984. *Case Study Research: Design and Methods.* Beverly Hills, Calif: Sage